

Deep Multi-Scale Mesh Feature Learning for Automated Labeling of Raw Dental Surfaces From 3D Intraoral Scanners

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Abstract—Precisely labeling teeth on digitalized 3D dental surface models is the precondition for tooth position rearrangements in orthodontic treatment planning. However, it is a challenging task primarily due to the abnormal and varying appearance of patients' teeth. The emerging utilization of intraoral scanners (IOSs) in clinics further increases the difficulty in automated tooth labeling, as the raw surfaces acquired by IOS are typically low-quality at gingival and deep intraoral regions. In recent years, some pioneering end-to-end methods (e.g., PointNet) have been proposed in the communities of computer vision and graphics to consume directly raw surface for 3D shape segmentation. Although these methods are potentially applicable to our task, most of them fail to capture fine-grained local geometric context that is critical to the identification of small teeth with varying shapes and appearances. In this paper, we propose an end-to-end deep-learning method, called *MeshSegNet*, for automated tooth labeling on raw dental surfaces. Using multiple raw surface attributes as inputs, *MeshSegNet* integrates a series of graph-constrained learning modules along its forward path to hierarchically extract multi-scale local contextual features. Then, a dense fusion strategy is applied to combine local-to-global geometric features for the learning of higher-level features for mesh cell annotation. The predictions produced by our *MeshSegNet* are further post-processed by a graph-cut refinement step for final segmentation. We evaluated *MeshSegNet* using a real-patient dataset consisting of raw maxillary surfaces acquired by 3D IOS. Experimental results, performed 5-fold cross-validation, demonstrate that *MeshSegNet* significantly outperforms state-of-the-art deep learning methods for 3D shape segmentation.

Index Terms—3D shape segmentation, geometric deep learning, automated tooth labeling, orthodontic treatment planning, 3D intraoral scanners.

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I. INTRODUCTION

In the clinical practice of dentistry, 3D dental surface models are routinely digitalized by scanning physical impressions (i.e., plaster models) of dental arches for orthodontic treatment planning, which can be an uncomfortable experience for patients (e.g., young children) and may even cause allergies due to some specific constituents of impression materials [1]. Compared with traditional hard-copy impressions, advanced intraoral scanners (IOSs) directly reconstruct the digital surface model of the dentition [2], and thus are becoming widespread because of their efficiency and safety. Precisely labeling teeth on digitalized 3D dental surfaces is important for the analyses and rearrangements of tooth positions [3]. However, developing automated methods for this purpose is challenging, since 1) the shapes of teeth vary dramatically across subjects and positions, and 2) the teeth of patients usually exhibit abnormal appearances affected by misalignments and crowding [4]. The utilization of 3D IOSs further increases the difficulty in tooth labeling, as the non-tooth parts (e.g., gingival tissues) of the raw surfaces usually present irregular shapes and the deep intraoral regions (e.g., the 2nd/3rd molars) may not be completely captured by the optical scanner.

Some methods have been proposed to annotate teeth on high-quality 3D dental models, instead of raw surfaces scanned by IOS. Those methods are direct and intuitive. For example, they may project 3D surfaces onto 2D images to perform segmentation [3] or predefine some general criteria to separate 3D mesh cells [5]. Consequently, they are typically not fully automatic, often requiring significant interactions even from experienced operators. Their performance is very sensitive to variation in tooth appearances [4].

Shape and geometry analysis using (deep) learning techniques, broadly investigated in computer vision and computer graphics communities, can be applied for tooth labeling on dental surfaces. For example, following [6], a multi-stage framework was proposed in [4] to reshape cell-wise hand-crafted features as 2D image patches for training convolutional neural networks (CNNs) for the classification of mesh cells on dental surfaces. Although straightforward, [6] and [4] improperly assumed that the input hand-crafted features are locally correlated to be an image, which ignores the unordered nature

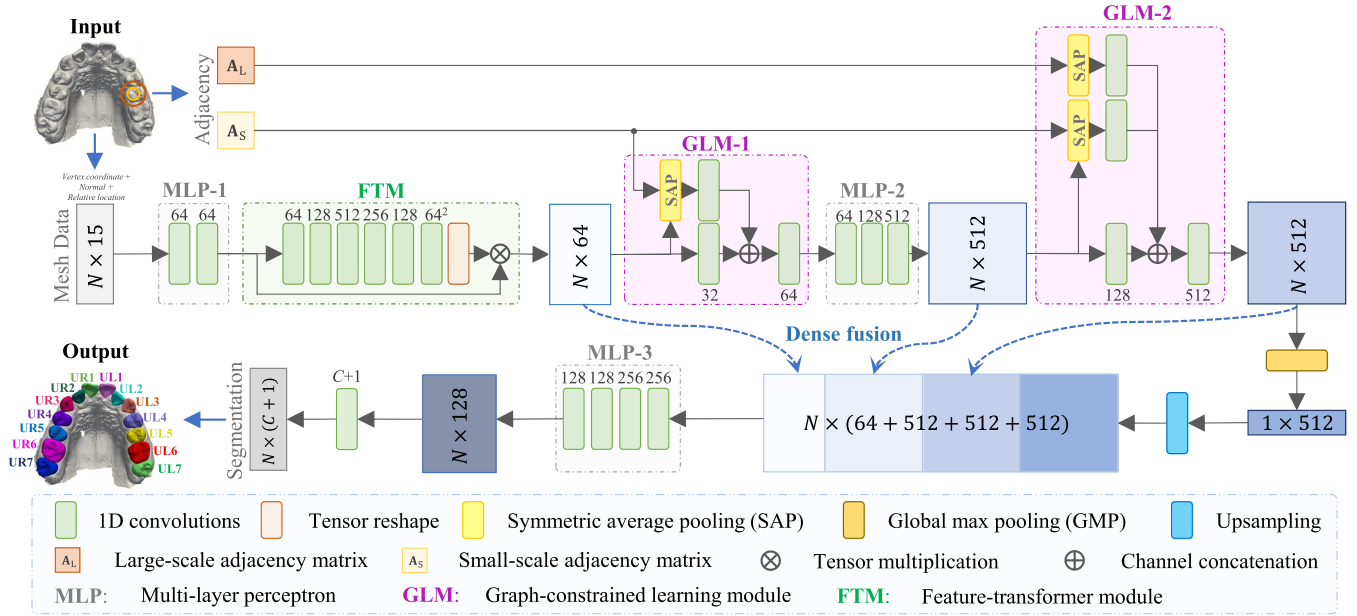


Fig. 1. Schematic diagram of MeshSegNet for automated tooth labeling on raw 3D dental surfaces acquired by IOS. The network takes as input raw mesh data (i.e., coordinates of cell vertices, cell normals and relative locations with respect to the whole surface) as input to learn hierarchically multi-scale contextual feature with graph-constrained learning modules (GLMs), and then adopts a dense fusion strategy to combine local-to-global geometric features for the labeling of individual teeth.

of those features, as reshaping them in different ways definitely produce different and irrelevant images. Also, the multi-stage framework proposed in [4] involves multiple (>6) isolated steps requiring considerable tuning parameters, and thus is hard to be applied to different datasets.

A novel deep-learning method, called PointNet [7], was proposed to learn translation-invariant deep geometric features directly from the *raw* surface data (e.g., mesh points or cells). While showing successful applications in both point- and surface-wise classification tasks, it is agnostic to capture local geometrical contexts that are indispensable for fine-grained segmentation. Some state-of-the-art methods, e.g., PointNet++ [8], recurrent slice network (RSNet) [9], and PointConv [10], have been proposed to extend PointNet by explicitly modeling local contexts. However, since most of these methods typically coarsely grouping points or cells into clusters based on spatial distances, their performance is limited in our specific task of fine-grained tooth segmentation, considering that each tooth is very small compared with the whole dental surface.

In this paper, we propose to directly learn high-level geometric features of the *raw dental surfaces* acquired by 3D IOS for automated tooth labeling. We propose an *end-to-end* deep-learning method called MeshSegNet. As shown in the schematic diagram in Fig. 1, MeshSegNet is an extension of PointNet. *First*, instead of solely using the coordinates of a single point, we adopt as the primary network input the coordinates of three cell vertices, as they naturally unite adjacent cells with shared vertices. This implicitly accounts for the spatial dependency of cells. Besides, we also include the normal and relative positions of cells with respect to the whole surface as additional inputs for more comprehensive cell description. *Second*, we propose *multi-scale graph-constrained*

learning modules and integrate them along the forward path of MeshSegNet, mimicking CNNs to hierarchically model multi-scale local geometric contexts on surfaces. *Third*, we design a dense fusion strategy that adopts dense skip connections to combine local-to-global features (i.e., cell-wise features, multi-scale contextual features, and translation-invariant holistic features), based on which higher-level features are extracted for cell annotation. *Fourth*, after the end-to-end prediction by our MeshSegNet, an efficient post-processing step, based on graph cut [11] and basic geometric properties of dental surface [4], is applied to further refining the tooth labeling results.

This work is an extension of a preliminary conference publication [12]. In addition to more detailed literature review, other major extensions in this journal paper include 1) a graph-cut post-processing step is included to further refine MeshSegNet predictions, 2) more real-patient dental surfaces acquired by IOS are included for the evaluation of segmentation performance, 3) MeshSegNet is compared with more state-of-the-art deep networks that explicitly model surface local dependencies on the enlarged dataset under different scenarios, and 4) more ablation studies are performed to comprehensively evaluate the effectiveness of different key modules of MeshSegNet.

The rest of the paper is organized as follows. Related work, including learning-based tooth labeling methods, PointNet and its variants, are briefly reviewed in Section II. The evaluation data and proposed method are described in Section III. Section IV presents the experimental settings and comparison results between different methods. In Section V, we discuss the effectiveness of different modules of our method via detailed ablation studies and then briefly summarize the potential limitations of our work. The paper is concluded in Section VI.

II. RELATED WORK

A. Tooth Labeling on Dental Surface Model

Conventional methods usually predefine geometric criteria to separate teeth from dental surface model. For example, Zou *et al.* [5] employed harmonic fields defined on triangular meshes to iteratively annotate teeth on dental surface model. Based on mesh curvatures, Kumar *et al.* [13] first separated teeth from gum, and then further separated individual teeth. Wu *et al.* [14] defined morphological skeleton for scanned dental meshes, based on which teeth were iteratively separated using region growing operations. Kondo *et al.* [3] projected 3D dental model onto two 2D images (i.e., a plan-view range image and a panoramic range image), from which teeth were segmented and reconstructed back to 3D space. Although these methods are intuitive, they typically rely on expert prior knowledge and tedious human interactions, causing sensitivity to variations in surface appearances.

Recently, a number of deep learning-based methods have been proposed for dental surface labeling in a data-driven fashion. For example, Xu *et al.* [4] proposed a multi-stage framework that integrates CNNs for tooth segmentation on 3D dental surface. For each mesh cell, they extracted and reshaped hand-crafted geometric features as an image patch to train CNN models for cell classification. The trained CNNs models were integrated into a complex pipeline integrating multiple steps, such as boundary-aware simplification, fuzzy clustering, and boundary smoothing, to hierarchically annotate individual teeth. Although this method achieved promising performance on high-quality surface data, the complex pipeline hampers its practical utility on other datasets, e.g., raw data acquired by IOS. Similar to [4], a multi-stage deep-learning framework was proposed by Tian *et al.* [15], preprocessing surface data with a sparse octree partitioning to construct 3D CNNs for hierarchical labeling of individual teeth.

B. 3D Shape Segmentation

Diverse end-to-end learning methods have been proposed in the computer vision and computer graphics communities for 3D shape classification or segmentation. These methods voxelize surfaces as volumes to construct 3D CNNs [16], [17] or/and render surfaces as multi-view images to construct 2D CNNs [18], [19], which inevitably result in transformation artifacts and information loss.

Directly using the raw surface data (e.g., vertex/cell positions) as input, PointNet [7] was proposed to learn translation-invariant deep geometric features with successive multi-layer perceptrons (shared across vertices) and global max pooling. Although achieving promising results in multiple tasks, PointNet processes different vertices/cells independently, ignoring their spatial dependency. To improve the performance of PointNet, local geometric contexts can be explicitly modeled. For example, PointNet++ [8] hierarchically groups points as increasingly large groups, and applies PointNets to learn group-wise geometric features. RSNet [9] sorts points/cells as ordered sequences from different directions (i.e., x -axis, y -axis, and z -axis), and then applies recurrent neural network (RNN) [20] to capture local dependencies.

PointCNN [21] learns χ -transformations to reorganize neighboring vertices/cells for general convolutional operations, which has then been successfully applied to automatically labeling dental surfaces [22]. Considering that the learned χ -transformations in PointCNN are not permutation-invariant, a more recent method called PointConv [10] was proposed to learn directly translation- and permutation-invariant convolution kernels on vertices/cells, showing improved performance.

Although these existing extensions of PointNet largely improve both surface classification and segmentation performance, they have several common limitations: 1) overly coarse modeling of local dependencies, and 2) using solely vertex/cell positions as inputs, ignoring other supplementary information that can be easily computed from raw surfaces. Notably, an interesting method called MeshNet [23] was proposed recently to tackle these limitations. However, the architecture of MeshNet is designed solely for surface-level classification, which is hard to be applied for dense classification tasks, such as our task of fine-grained labeling of raw dental surfaces.

III. MATERIALS AND METHOD

A. Data and Pre-Processing

The studied dataset consists of raw maxillary dental surfaces that were directly acquired and reconstructed by an 3D intraoral scanner (i.e., iTero Element[®]) for 30 different subjects. Each original surface roughly contains 100,000 mesh cells/triangles. They were downsampled to 10,000 cells with the original topology preserved. On the downsampled surface, the ground-truth segmentations for $C = 14$ teeth, from the central incisor to the second molar on both right and left sides (denoted as UR1-UR7 and UL1-UL7, respectively), were annotated by a resident guided by experienced dentists.

In MeshSegNet, each cell on the dental surface is described by a 15-D input vector. Specifically, rather than the vertex/cell coordinates used in PointNet [7] and other state-of-the-art methods [8]–[10], we adopt the coordinates of cell vertices (9 elements) as the primary input. This formulation naturally considers the spatial connections between neighboring vertices and cells. In addition, the normal vector (3 elements) and the relative position (3 elements) of each cell with respect to the whole surface are also included as additional inputs to provide supplementary information for cell-wise labeling. The relative position of a cell is simply defined as its centroid normalized along each axis in terms of the maximum and minimum coordinates for all cells on the surface. As the input of MeshSegNet, the feature vectors for different cells are finally z -score normalized along each dimension.

B. MeshSegNet

We assume that the input raw surface is an $N \times 15$ matrix $\mathbf{F}^0$, where N is the number of mesh cells. MeshSegNet applies the architecture shown in Fig. 1 to establish the nonlinear mapping from $\mathbf{F}^0$ to $\mathbf{P}$, an $N \times (C + 1)$ matrix with each row denoting the probabilities of the respective cell belonging to different classes (i.e., UR1-UR7, UL1-UL7, and gingiva). Specifically, a series of multi-layer perceptrons (MLPs) is used in MeshSegNet to extract increasingly higher-level geometric

features for each mesh cell. The learnable parameters of each MLP are shared across all cells, akin to convolutional layers that slide a window on pixel/voxel grids in CNNs.

Inspired by [7], the first MLP (i.e., MLP-1) in our MeshSegNet is followed by a feature-transformer module (FTM). The FTM is a special MLP, which predicts a transformation matrix $\mathbf{T} \in \mathbb{R}^{64 \times 64}$ from $\mathbf{F}^1 \in \mathbb{R}^{N \times 64}$ (i.e., the feature presentations produced by MLP-1), based on which the initial features for all mesh cells are jointly updated as

$$\hat{\mathbf{F}}^1 = \mathbf{F}^1 \mathbf{T}. \quad (1)$$

That is, the FTM aligns the initial features for different cells onto a canonical feature space, facilitating the learning of more representative and robust features in subsequent layers [7]. Compared with PointNet, our MeshSegNet removes the unnecessary spatial transformer (between the input and MLP-1) to control network complexity, considering that the input raw features are cell-wise (i.e., not point coordinates) in this study.

Despite the similarity in the usage of MLPs and FTM, MeshSegNet presents two key innovations compared with PointNet, including 1) graph-constrained hierarchical learning modules for explicitly learning multi-scale local geometric features, and 2) a dense fusion strategy for combining local-to-global (i.e., cell-level, multi-scale contextual, and surface-level holistic) features for more comprehensive cell description.

1) Multi-Scale Graph-Constrained Learning: We propose graph-constrained learning modules (GLMs) to explicitly model local geometric context on the input surface data. As shown in Fig. 1, these GLMs (i.e., GLM-1 and GLM-2) are implemented with increasing scales and are integrated at different stages along the forward path of our network (i.e., after the FTM and MLP-2, respectively). MeshSegNet mimics CNNs in gradually increasing the receptive field for learning hierarchically multi-scale contextual features.

To construct GLM-1 and GLM-2, we regard each cell as the centroid to define two 3D balls with different radii (i.e., $0 < r_S < r_L$). For each centroid, the cells located within the respective balls are selected as its neighbors, based on which two $N \times N$ adjacency matrices (i.e., $\mathbf{A}_S$ and $\mathbf{A}_L$ for r_S -ball and r_L -ball, respectively) are constructed for the input mesh data, with each element describing the graph connections between the corresponding two cells (1 for connected and 0 otherwise) in the underlying Euclidean space. Considering that the input feature matrix $\mathbf{F}^0$ is z -score normalized for each surface, we empirically set $r_S = 0.1$ and $r_L = 0.2$ in our experiments.

Based on $\mathbf{A}_S$, GLM-1 first applies the *symmetric average pooling* (SAP), a graph-based fusion operation, on $\hat{\mathbf{F}}^1$ (i.e., the output of FTM) to propagate the local contextual information onto each centroid cell. Assume $\tilde{\mathbf{A}}_S = \mathbf{A}_S + \mathbf{I}$ is the adjacency with self-loops, and $\tilde{\mathbf{D}}_S^{-\frac{1}{2}} \tilde{\mathbf{A}}_S \tilde{\mathbf{D}}_S^{-\frac{1}{2}}$ is the corresponding symmetric-normalized adjacency, where $\tilde{\mathbf{D}}_S$ denotes the diagonal degree matrix. The SAP operation in terms of $\hat{\mathbf{F}}^1$ given $\mathbf{A}_S$ is defined as

$$\tilde{\mathbf{F}}^1 = \text{f}_{\text{SAP}}(\hat{\mathbf{F}}^1 | \mathbf{A}_S) = \left(\tilde{\mathbf{D}}_S^{-\frac{1}{2}} \tilde{\mathbf{A}}_S \tilde{\mathbf{D}}_S^{-\frac{1}{2}} \right) \hat{\mathbf{F}}^1, \quad (2)$$

where $\tilde{\mathbf{F}}^1 \in \mathbb{R}^{N \times 64}$ is the updated cell-wise feature matrix encoding local geometric contexts. After SAP, both $\tilde{\mathbf{F}}^1$ and $\hat{\mathbf{F}}^1$ are further squeezed by the shared-weights 1D Convs with 32 channels. The resulting feature matrices are then concatenated across channels, followed by the fusion by another 1D Conv with 64 channels.

Different from GLM-1, GLM-2 enlarges the receptive field and learns multi-scale contextual features. Specifically, based on Eq. (2), the $N \times 512$ feature matrix from MLP-2 (i.e., $\mathbf{F}^2$) is processed by two parallel SAPs given $\mathbf{A}_S$ and $\mathbf{A}_L$, respectively. The resulting feature matrices, i.e., $\text{f}_{\text{SAP}}(\mathbf{F}^2 | \mathbf{A}_S)$ and $\text{f}_{\text{SAP}}(\mathbf{F}^2 | \mathbf{A}_L)$, and $\mathbf{F}^2$ are then squeezed by the shared-weights 1D Convs with 128 channels, which are finally concatenated across channels and fused by another 1D Conv with 512 channels.

Notably, the proposed GLM can be regarded as an extension of graph convolutional networks (GCNs) [24], [25]. For example, the complete operations in GLM-1 output a feature matrix $\mathbf{F}^{S1}$ with the form of

$$\mathbf{F}^{S1} = \sigma \left(\left\{ \sigma(\hat{\mathbf{F}}^1 \mathbf{W}^1) \oplus \sigma(\tilde{\mathbf{F}}^1 \mathbf{W}^1) \right\} \mathbf{W}^2 \right), \quad (3)$$

where $\mathbf{W}^1$ and $\mathbf{W}^2$ are the learnable weights for 1D Convs with 32 and 64 channels, respectively, and $\oplus$ denotes the channel-wise concatenation of feature matrices. Let $\sigma(\cdot)$ be the ReLU activation, then we have

$$\sigma(\tilde{\mathbf{F}}^1 \mathbf{W}^1) = \sigma(\tilde{\mathbf{D}}_S^{-\frac{1}{2}} \tilde{\mathbf{A}}_S \tilde{\mathbf{D}}_S^{-\frac{1}{2}} \hat{\mathbf{F}}^1 \mathbf{W}^1) \quad (4)$$

in (3) being similar to graph Conv layers [24]. The difference is that the graph connections used in our GLMs are dynamic, since different input surfaces have their own definitions of cell-wise neighborhood, i.e., the adjacency $\mathbf{A}_S$ is input-specific. This property could make our GLM be more flexible than spectrum GCNs requiring static adjacencies.

2) Dense Fusion of Local-to-Global Features: We apply global max pooling (GMP) on the output of GLM-2 to produce the translation-invariant holistic features, which encode the semantic information of the whole input surface. In contrast to PointNet that combine only cell/point-wise and holistic features with skip connections, we assume that the multi-scale contextual features (produced by the intermediate GLMs) could provide additional information to more comprehensively describe mesh cells. Correspondingly, we densely concatenate the local-to-global features from FTM, GLM-1, GLM-2 and (upsampled) GMP, which are further processed by MLP-3 to yield a $N \times 128$ feature matrix. Based on this matrix, a 1D Conv layer with softmax activation is used to predict a $N \times (C + 1)$ probability matrix $\mathbf{P}$, with each row denoting the probabilities of the respective cell belonging to specific categories (i.e., C teeth and gingiva).

C. Implementation

As shown in Fig. 1, the MeshSegNet architecture consists of three MLPs (i.e., MLP-1, MLP-2, and MLP-3). MLP-1 contains two 1D Convs, both with 64 channels. MLP-2 has three 1D Convs, each with 64, 128, and 512 channels, respectively. MLP-3 contains four 1D Convs, each with 256, 256, 128, and 128 channels, respectively. Apart from these MLPs, our

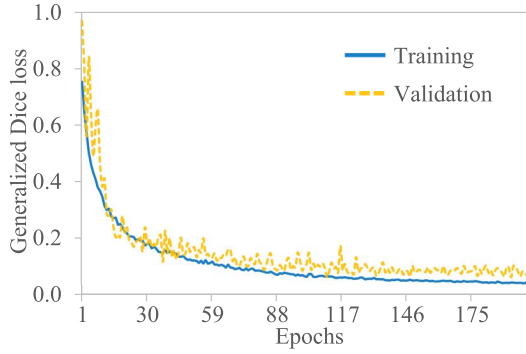


Fig. 2. The training and validation logs of our MeshSegNet in one iteration of the 5-fold cross-validation on the studied dataset.

MeshSegNet also includes one FTM, two GLMs (i.e., GLM-1 and GLM-2), and a final 1D Conv layer to output softmax segmentation probabilities. All 1D Convs in these MLPs and the GLMs are followed by batch normalization (BN) and ReLU activation. To learn the 64×64 feature transformation matrix $\mathbf{T}$ for $\mathbf{F}^1$ (i.e., the output of MLP-1), FTM employs six 1D Convs with 64, 128, 512, 256, 128, and 64^2 channels, respectively, where each of the first five layers is followed by BN and ReLU, while the last layer (without BN and ReLU) is followed by a tensor reshape operation.

1) Training & Data Augmentation: The model was trained by minimizing the generalized Dice loss [26] using the AMSGrad variant [27] of the Adam optimizer. To improve the generalization ability, we augmented the training and validation sets by combining 1) random rotation, 2) random translation, and 3) random rescaling (e.g., zoom-in/out) of each 3D surface in reasonable ranges. Specifically, along each axis in the 3D space, a training/validation surface has 50% probability to be translated with a displacement and zoomed with a ratio uniformly sampled between $[-20, 20]$ and $[0.7, 1.3]$, respectively. Also, each training/validation surface has 50% probability to be x -axis and y -axis rotated with angles uniformly sampled between $[-\frac{\pi}{4}, \frac{\pi}{4}]$, respectively, and z -axis rotated with an angle uniformly sampled between $[-\pi, \pi]$. The combination of these random operations simulated 100 “new” cases from each original surface. After that, on each training/validation surface (with roughly 10,000 cells), we randomly sampled 50% cells from each tooth and then randomly sampled the remaining cells from the gingival to form the input during training (with 6,000 cells in total). In this way, the training set was largely enriched, and the imbalanced learning challenge, caused by the fact that each tooth only covers a very small part on the dental surface, was properly mitigated. With the mini-batch size setting as 10, the network (~ 1.7 million learnable parameters) was trained for 200 epochs in total. Each epoch took around 45 mins on a PC with 64 Gb memory and a 11 Gb graphics card (i.e., GeForce GTX 1080 Ti). The training and validation logs plotted in Fig. 2 show that the trained network generalizes well on our studied data.

2) Inference: We directly applied our trained MeshSegNet model on unseen test surfaces to predict the corresponding probability matrices. That is, similar to fully convolutional networks for image segmentation [28], [29], our MeshSegNet

can directly process the whole dental surfaces with varying sizes in the test phase, which is practically useful.

D. Post-Processing With Graph Cut

The segmentations produced by deep neural networks (e.g., our MeshSegNet) may contain isolated false predictions or non-smooth boundaries. Graph-based post-processing (e.g., graph cuts [11]) has been successfully applied for the refinement of the results produced by automated methods in both the tasks of image segmentation [30]–[32] and surface labeling [4], [6], [33], [34].

We perform graph-cut on the segmentation results produced by MeshSegNet for the whole dental surface. Assume $\mathcal{S} = \{s_i\}_{i=1}^N$ is a test surface consisting of N mesh cells, and the corresponding predictions from our MeshSegNet form a $N \times (C + 1)$ probability matrix $\mathbf{P}$, with $\mathbf{P}_i$ ($i = 1, \dots, N$) denoting the class probabilities for cell s_i . We define the spatial neighborhood of s_i as $\mathcal{N}_i$, which consists of adjacent cells that share edges with s_i . Then, the final label map $L = \{L_i\}_{i=1}^N$ (with $L_i \in \{1, \dots, C + 1\}$) for $\mathcal{S}$ is determined by multi-label graph cut [11]:

$$\arg \min_L \sum_{i=1}^N -\log(\mathbf{P}_{i,L_i} + \epsilon) + \lambda \sum_{i=1}^N \sum_{i^* \in \mathcal{N}_i} \mathcal{C}(L_i, L_{i^*}), \quad (5)$$

where the first term is the data fitting term, the second term defines local smoothness in the label space, ϵ is a small scalar (i.e., $1e-4$) to ensure numerical stability, and λ is a tuning parameter that balances the contributions of two terms. In our experiments, λ was empirically set as 20 as it led to relatively good results on a validation set. Considering that the boundaries between teeth and gingiva on dental surfaces are typically concave, we follow [4] to define the local label consistency, i.e., $\mathcal{C}(L_i, L_{i^*})$, in the second term of (5) as

$$\mathcal{C}(L_i, L_{i^*}) = (1 - \delta(L_i, L_{i^*})) \times \psi(s_i, s_{i^*}), \quad (6)$$

where $\delta(L_i, L_{i^*})$ is the Dirac delta function, which equals 1 iff $L_i = L_{i^*}$. On the other hand, $\psi(s_i, s_{i^*})$ favors concave angles, such as

$$\psi(s_i, s_{i^*}) = -\gamma_{i,i^*} \times \log\left(\frac{\theta_{i,i^*}}{\pi}\right) \times d_{i,i^*}, \quad (7)$$

where θ_{i,i^*} is the angle between s_i and s_{i^*} ; $\gamma_{i,i^*} = 1$ (or $1 + |n_i \cdot n_{i^*}|$) when θ_{i,i^*} is concave (or not), with n_i and n_{i^*} denote the normals of s_i and s_{i^*} , respectively; and d_{i,i^*} is the distance between cell s_i and its neighbor s_{i^*} . Following [4], d_{i,i^*} was implemented as the Euclidean distance in our experiments for computational simplicity, although geodesic distance [35] could also be a promising alternative.

IV. EXPERIMENTS

A. Competing Methods

Using the dataset introduced in Section III-A, MeshSegNet was compared with **PointNet** [7], **PointNet++** [8], and **PointConv** [10], for automated labeling of 14 teeth on the maxillary dental surfaces. All these competing methods are closely related to MeshSegNet and directly utilize the raw

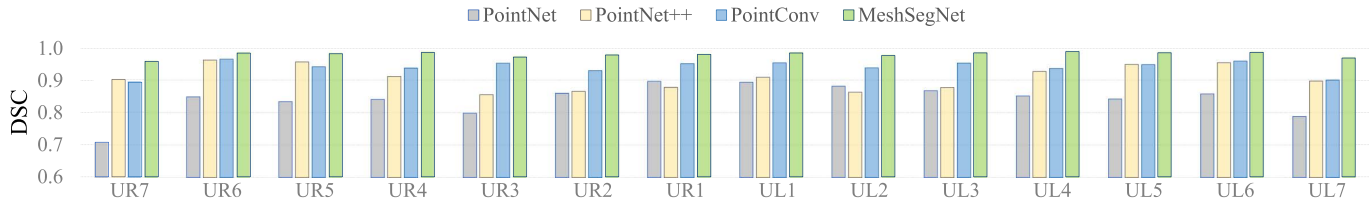


Fig. 3. Segmentation results in terms of DSC for each of the 14 teeth (i.e., UR1-UR7 and UL1-UL7) quantified under 5-fold cross-validation for three state-of-the-art competing methods (i.e., PointNet, PointNet++, and PointConv) and our MeshSegNet method.

surface inputs (i.e., coordinates and normals) to learn high-level geometric features for 3D shape segmentation. They are briefly summarized below.

- 1) **PointNet**: The architecture of PointNet was implemented following the original paper [7]. The input was $N \times 3$ matrix, where each row denotes the 3D coordinates of a mesh cell and N is the total number of cells. We trained PointNet and our MeshSegNet in the same way. That is, following the strategies described in Section III-C, $N = 6,000$ cells were sampled for training, and the trained network processed directly the whole dental surface ($N \approx 10,000$ cells) to output an $N \times 15$ matrix, where each row denotes the probabilities of a specific cell belonging to each tooth and the gingiva tissue. With the mini-batch size setting as 10, we trained PointNet for 200 epochs.
- 2) **PointNet++**: This method is an intuitive extension of PointNet by hierarchically modeling the spatial relationship between neighboring points or mesh cells. The architecture of the PointNet++ used in this paper was similar to the original paper [8], except that we doubled the numbers of local groups generated in the first two sampling layers for more fine-grained segmentation, considering that in our task each tooth is very small compared with the gingival tissue. The training and test samples for PointNet++ were defined in the same way as that for PointNet. The mini-batch size was set to 10 during training, and the network was trained 2,000 epochs, as we experimentally observed that PointNet++ converged slower than the original PointNet.
- 3) **PointConv**: The PointConv method employs translation-invariant and permutation-invariant (de)convolutions on non-uniform surface data [10], based on which an encoder-decoder network was designed for 3D shape segmentation. The architecture of PointConv in this paper was kept similar to that used in the original paper [10]. The training and test samples were prepared in the same way as that for other competing methods. The network was trained 200 epochs, and the mini-batch size was set to 5 due to GPU memory limitation.

Notably, in line with MeshSegNet, all these competing methods were followed by the same graph-cut post-processing step (i.e., Section III-D) for segmentation refinement. Also, for the fair comparisons, all networks were trained using the same loss function (i.e., the generalized Dice loss [26]) and the training set was augmented in the same way described in Section III-C1.

TABLE I

SEGMENTATION RESULTS (MEAN $\pm$ STANDARD DEVIATION) QUANTIFIED UNDER 5-FOLD CROSS-VALIDATION FOR THREE STATE-OF-THE-ART DEEP LEARNING METHODS AND OUR METHOD. ALL METHODS WERE FOLLOWED BY THE SAME GRAPH-CUT POST-PROCESSING. SYMBOL * INDICATES THAT OUR METHOD SIGNIFICANTLY OUTPERFORMED THE RESPECTIVE COMPETING METHOD (WITH $P < 1.5E-9$)

Metric	PointNet [7]	PointNet++ [8]	PointConv [10]	MeshSegNet
DSC	0.840 $\pm$ 0.117*	0.907 $\pm$ 0.067*	0.939 $\pm$ 0.060*	0.981 $\pm$ 0.028
SEN	0.910 $\pm$ 0.129*	0.988 $\pm$ 0.037	0.983 $\pm$ 0.047	0.989 $\pm$ 0.035
PPV	0.795 $\pm$ 0.132*	0.846 $\pm$ 0.107*	0.904 $\pm$ 0.086*	0.975 $\pm$ 0.034

B. Experimental Setup

The segmentation performance was evaluated via 5-fold cross-validation. In each iteration, one surface or subject was randomly split from the training set for validation, and the resulting training and validation sets were then enlarged using the data augmentation protocol described in Section III-C1, by simulating 100 “new” surfaces from each training/validation surface. The training/validation inputs with 6,000 mesh cells were then randomly sampled on each surface on-the-fly during training. Using the manual annotations as the ground truth, the results for the automated methods were quantitatively evaluated by three metrics, i.e., 1) the Dice similarity coefficient (DSC), 2) the sensitivity (SEN), and 3) the positive prediction value (PPV), whose values are positively proportional to the segmentation performance.

C. Results

1) **Quantitative Evaluations**: The overall segmentation results in terms of the three metrics are summarized in Table I, from which we can have two main observations. *First*, compared with PointNet, the other three methods (i.e., PointNet++, PointConv, and MeshSegNet) yield better results. For example, MeshSegNet leads to 14.1%, 7.9%, and 18.0% improvements of DSC, SEN, and PPV, respectively, which are all statistically significant with $p < 1.5e-9$. This verifies that explicitly modeling local geometric context (as in PointNet++, PointConv, and our MeshSegNet) is desired for more precise segmentation on 3D surfaces. *Second*, compared with PointNet++ and PointConv, MeshSegNet results in consistently better performance for all metrics. For example, MeshSegNet significantly outperforms (with $p < 1.5e-9$) PointConv in terms of both DSC and PPV. It implies that our method can more precisely

label small objects in the specific task of tooth segmentation, e.g., identifying the boundaries between adjacent teeth (less false positives or larger PPV value), suggesting the effectiveness of the proposed multi-scale graph-constrained learning modules and the local-to-global dense fusion strategy.

We also quantified the per-tooth segmentation results obtained by different methods, and the respective DSC values are summarized in Fig. 3. From Fig. 3 we can observe that, in consistent with the overall results presented in Table I, MeshSegNet robustly led to better performance than other competing methods in labeling all teeth (i.e., UR1 to UR7 and UL1 to UL7). On the other hand, it is worth noting that the improvement brought by MeshSegNet is more significant for the segmentation of the *molars* (i.e., UR7 and UL7). For example, compared with the state-of-the-art PointConv, MeshSegNet improves the DSC values from 0.894 to 0.959 ($p < 6e-3$) and 0.899 to 0.970 ($p < 4e-5$) for labeling UR7 and UL7, respectively. These results further suggest the effectiveness of MeshSegNet, especially considering that labeling molars is a very challenging task, as they locate at deep intraoral regions and might not be completely captured by the light source of IOS.

2) *Qualitative Evaluations*: We also visually compared the automated labeling results with representative examples presented in Fig. 4. From the first two rows of Fig. 4, we can observe that MeshSeNet results in overall better performance than the competing methods, especially in distinguishing the boundaries between adjacent teeth and in segmenting small teeth (as indicated by the yellow arrows). From the third and fourth rows of Fig. 4, we can see that MeshSegNet can also more effectively capture the incompletely scanned molars and crowded/misaligned incisors (as indicated by the yellow dotted circles). The visual evaluation results shown in Fig. 4 are consistent with the quantitative evaluations in Section IV-C1 (including Table I and Fig. 3), which further suggest the practical utility of MeshSegNet for automated tooth labeling on raw dental surfaces.

V. DISCUSSION

In this section, we will first perform ablation studies to evaluate the effectiveness of the important components of MeshSegNet. We will also point out some limitations of MeshSegNet and the potential solutions.

A. Effectiveness of Graph-Cut Post-Processing

As described in Section III-D, the automated segmentations produced by a deep network can be further refined by a specific graph-cut (GC) post-processing step. To verify the effectiveness of this GC refinement step, we applied it after each competing method, with the quantitative segmentation results summarized in Fig. 5 and a representative example shown in Fig. 6, respectively.

From Fig. 5, we can draw at least two observations. *First*, the GC refinement step consistently improved the performance of all deep-learning methods in terms of all metrics, which verifies its effectiveness in segmentation post-processing. *Second*, it is worth noting that, compared with other methods,

TABLE II

THE 5-FOLD CROSS-VALIDATION RESULTS (IN TERMS OF DSC, SEN, AND PPV) OBTAINED BY OUR MESHSEGNET USING DIFFERENT INPUT COMBINATIONS, WHERE W/O NORMAL, W/O RELATIVE LOCATION, AND W/O VERTEX COORDINATE STAND FOR USING VERTEX COORDINATES + NORMAL, VERTEX COORDINATES + RELATIVE LOCATION, AND CELL COORDINATE + NORMAL + RELATIVE LOCATION, RESPECTIVELY. IN CONTRAST, WITH ALL INPUTS STANDS FOR USING VERTEX COORDINATES + NORMAL + RELATIVE LOCATION

Input	DSC	SEN	PPV
w/o normal	0.874 ± 0.060	0.864 ± 0.075	0.892 ± 0.084
w/o relative location	0.947 ± 0.068	0.955 ± 0.074	0.943 ± 0.075
w/o vertex coordinate	0.937 ± 0.087	0.943 ± 0.094	0.934 ± 0.092
with all inputs	0.952 ± 0.049	0.960 ± 0.053	0.948 ± 0.064

the improvements brought by MeshSegNet are much more significant before the GC refinement than thereafter. Specifically, it improves the average DSC values from 0.740, 0.826, and 0.890 to 0.952, when compared with PointNet ($p < 1e-20$), PointNet++ ($p < 1e-18$), and PointConv ($p < 1e-10$), respectively, which further justifies the effectiveness of MeshSegNet in 3D dental surface labeling.

The qualitative evaluations in Fig. 6 are consistent with the quantitative results presented in Fig. 5, as the GC refinement visually improved the segmentations from all methods and the differences between our method and other competing methods are more significant before the GC refinement.

B. Effectiveness of Combining Multiple Inputs

Instead of using solely the 3D coordinates of the points or mesh cells (as in PointNet, PointNet++, and PointConv), MeshSegNet adopts the vertex coordinates of each cell (9 units) as the primary input, considering that they naturally consider the spatial dependencies between linked cells. Besides, the normal (3 units) and relative location (3 units) of each cell are also included in our MeshSegNet as the additional input to provide supplementary information. In this group of experiments, we performed the ablation study to evaluate the effectiveness of jointly using multiple inputs. Specifically, we implemented the variants of our MeshSegNet using different input combinations, including 1) vertex coordinates + normal (i.e., w/o relative location), 2) vertex coordinates + relative location (i.e., w/o normal), and 3) cell coordinate + normal + relative location (i.e., w/o vertex coordinates). The segmentation results of those variants were then compared with that of the original MeshSegNet (i.e., with all inputs) under 5-fold cross-validation, such as shown in Table II.

From Table II, we can see that, compared with cell coordinate, using vertex coordinates instead (i.e., w/o vertex coordinate vs. with all input) improved the tooth labeling performance in terms of all metrics. It is potentially due to the reason that using vertex coordinates naturally unified adjacent cells, as they share at least one vertex. On the other hand, we can also see that using normal and relative location as the supplementary information could also help refine the tooth labeling performance (refers to w/o normal vs. with all inputs,

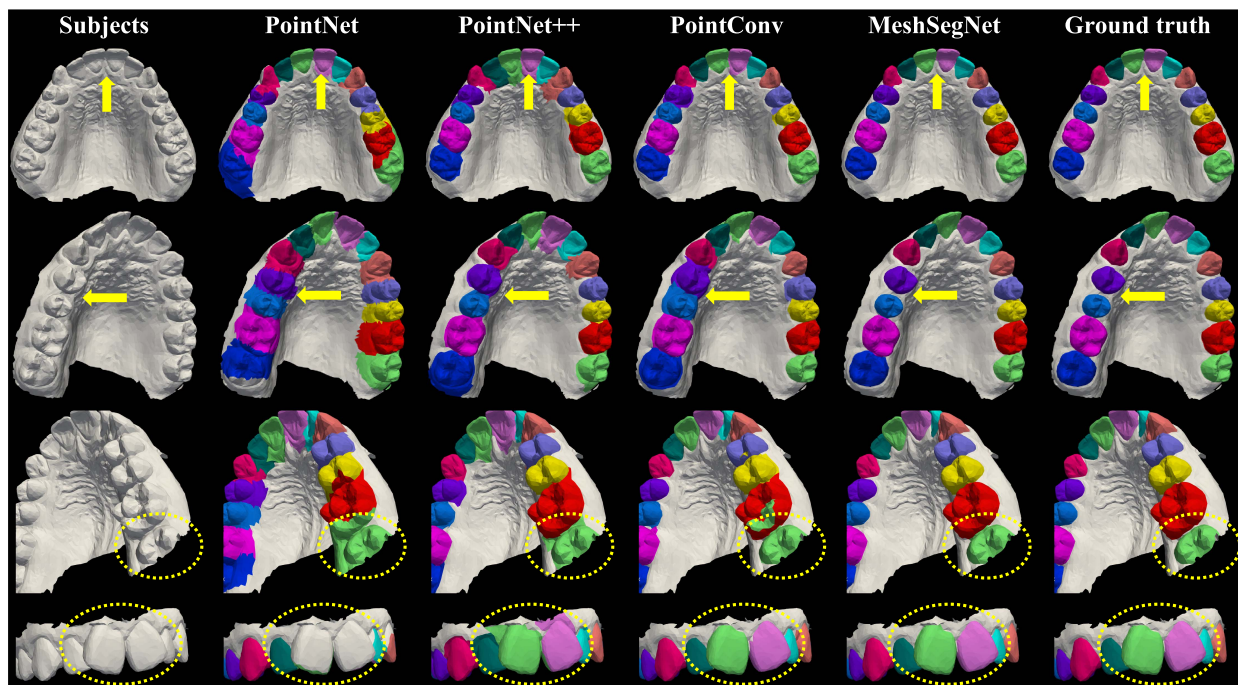


Fig. 4. Automated labeling produced by four different methods (followed by graph-cut-based post-processing) and the respective manual ground-truth annotations for four representative examples, with each row corresponding to a specific example. Different colors represent different teeth. Yellow arrows and dotted circles point out some superior segmentation details produced by our method compared with other competing methods.

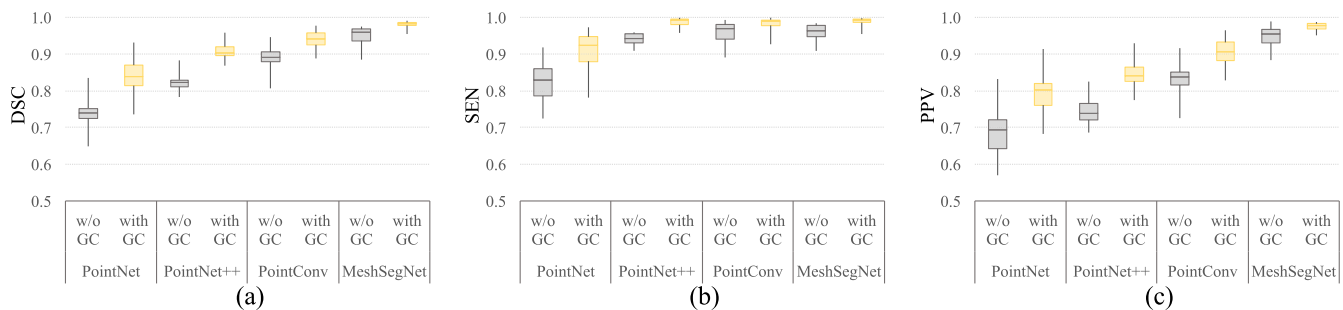


Fig. 5. The standard box plots of the segmentation results, in terms of (a) DSC, (b) SEN, and (c) PPV, obtained by different methods under 5-fold cross-validation. Here, w/o GC and with GC stand for the results yielded by the deep-learning methods without and with graph-cut post-processing, respectively.

and w/o relative location vs. with all inputs, respectively). Notably, based on the comparison between w/o normal and w/o relative location, we can see that normal brought larger performance improvements than relative position. One potential reason could be that normal describes a cell from a totally different view, and thus provides relatively more additional information for vertex coordinates. To analyze their influence in more detail, a segmentation example is shown in Fig. 7, from which we can see that normal and relative location may help refine the boundary between neighboring teeth (i.e., (a) vs. (c)) and remove isolated mislabeling (i.e., (b) vs. (c)), respectively.

Notably, including normal and relative location as supplementary information may also help refine the segmentations produced by other competing methods. For example, we implemented variants of PointNet by using the input combinations: 1) cell coordinate + normal, 2) cell coordinate

+ relative location, and 3) cell coordinate + normal + relative location, respectively. In Fig. 8, these variants are compared with the original PointNet, based on which we can see that including normal and relative location as supplementary features is also beneficial for PointNet. All these evaluations verify the effectiveness of using multiple complementary inputs in our proposed method.

C. Effectiveness of Multi-Scale Feature Learning

As described in Section III-B1, multi-scale graph-constrained learning modules (GLMs) were integrated in our MeshSegNet model to learn hierarchically geometric features at different scales for tooth labeling. In this group of experiments, we performed an ablation study to evaluate the effectiveness of our multi-scale GLMs. Specifically, two variants of our MeshSegNet were implemented, which adopted

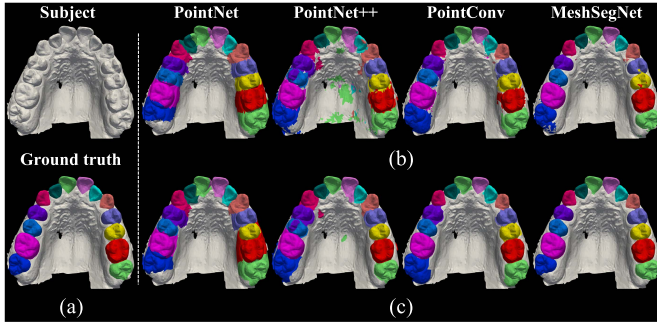


Fig. 6. A representative example of the segmentations produced by different methods with/without graph-cut (GC) refinement, where (a) presents the manual ground-truth, and (b) and (c) present the automated annotations without and with GC refinement, respectively.

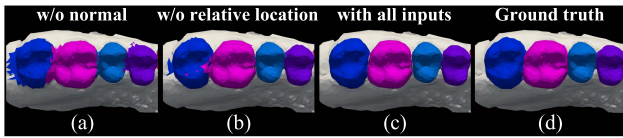


Fig. 7. A segmentation example for our MeshSegNet using different input combinations, where w/o normal and w/o relative location stand for using vertex coordinates + normal and vertex coordinates + relative location, respectively, while with all inputs denotes the original MeshSegNet.

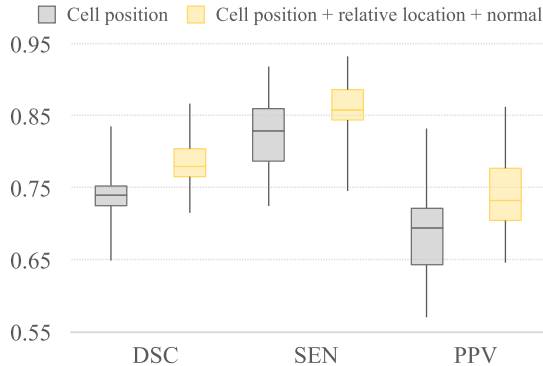


Fig. 8. The 5-fold cross validation results produced by PointNet using cell position and cell position + relative location + normal as inputs, respectively.

solely A_S and A_L in GLM-1 and GLM-2 of Fig. 1 to capture mono-scale local geometric context, respectively. Under 5-fold cross-validation, these two mono-scale variants (denoted as MeshSegNet- A_S and MeshSegNet- A_L , respectively) were compared with the original multi-scale version, with the quantitative results summarized in Table III. From this table, we can see that the multi-scale version outperformed the two mono-scale variants in terms of all metrics, which implies that the combination of multi-scale GLMs to hierarchically capture local geometric context is beneficial for the tooth labeling task.

To analyze the effectiveness in more detail, a segmentation example is presented in Fig. 9. It can be found that, compared with MeshSegNet, MeshSegNet- A_S under segmented each tooth (i.e., more false negatives), while MeshSegNet- A_L resulted in more mis-segmentation on boundaries. It suggests

TABLE III

THE SEGMENTATION RESULTS QUANTIFIED UNDER 5-FOLD CROSS-VALIDATION FOR THE MONO-SCALE VARIANTS AND THE ORIGINAL MULTI-SCALE VERSION OF OUR MESHSEGNET. HERE, MESHSEGNET- A_S AND MESHSEGNET- A_L STAND FOR USING SOLELY A_S AND A_L IN THE GLMS, RESPECTIVELY

Metric	Mono-scale learning		Multi-scale learning
	MeshSegNet- A_S	MeshSegNet- A_L	
DSC	0.918 ± 0.062	0.900 ± 0.109	0.952 ± 0.049
SEN	0.938 ± 0.075	0.898 ± 0.128	0.960 ± 0.053
PPV	0.908 ± 0.088	0.910 ± 0.106	0.948 ± 0.064

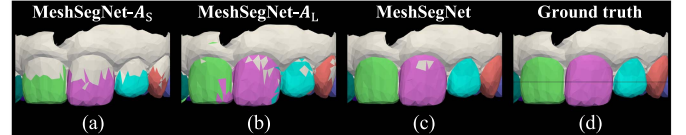


Fig. 9. A segmentation example for our method using mono-scale GLMs (i.e., MeshSegNet- A_S and MeshSegNet- A_L) and multi-scale GLMs (i.e., MeshSegNet), respectively.

that combining multi-scale modules can provide complementary information to each other for more precise tooth labeling.

D. Effectiveness of Dense Fusion Strategy

As described in Section III-B2, local-to-global features were densely connected in our MeshSegNet to learn higher-level features for the labeling of mesh cells on dental surfaces. In this group of experiments, we performed an ablation study to evaluate the effectiveness of this dense fusion strategy. Specifically, to form a variant of our MeshSegNet without dense fusion strategy (denoted as w/o dense fusion), we fused only the mesh-wise and holistic features for MLP-3, by removing the connections from GLM-1 and GLM-2 in Fig. 1. In addition, we also implemented a dense-fusion variant of PointNet, by concatenating the intermediate features with mesh-wise and holistic features in PointNet.

Using 5-fold cross-validation, the segmentation results produced by PointNet and our MeshSegNet with and without the dense-fusion strategy are summarized in Table IV. It can be seen that the dense fusion strategy significantly improves the performance for both methods, especially for our MeshSegNet. This is mainly because, in contrast to PointNet, the intermediate features in our MeshSegNet can provide multi-scale local contextual information that are complementary to cell-wise and holistic feature representations.

E. Limitations and Future Work

Although our MeshSegNet method achieved state-of-the-art performance in the challenging task of tooth labeling on raw dental surfaces, its practical usage is still hampered by several limitations. *First*, since the current training set only contains the maxillary dental surfaces with the complete 14 teeth, the respective MeshSegNet model cannot *robustly* handle special cases with extreme appearance. For example, in Fig. 10, we show the labeling results for two cases with

TABLE IV

THE SEGMENTATION RESULTS QUANTIFIED UNDER 5-FOLD CROSS-VALIDATION FOR POINTNET AND OUR MESHSEGNET WITH AND WITHOUT (I.E., W/O) THE DENSE FUSION STRATEGY

Method	Metric	w/o dense fusion	with dense fusion
PointNet	DSC	0.784 $\pm$ 0.111	0.826 $\pm$ 0.118
	SEN	0.860 $\pm$ 0.141	0.870 $\pm$ 0.136
	PPV	0.741 $\pm$ 0.136	0.802 $\pm$ 0.143
MeshSegNet	DSC	0.890 $\pm$ 0.093	0.952 $\pm$ 0.049
	SEN	0.904 $\pm$ 0.117	0.960 $\pm$ 0.053
	PPV	0.891 $\pm$ 0.113	0.948 $\pm$ 0.064

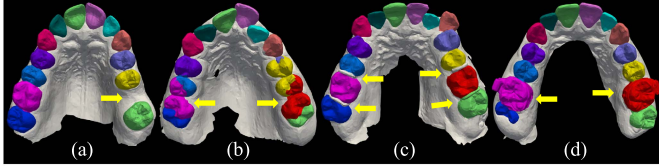


Fig. 10. Labeling results produced by our MeshSegNet for subjects with extreme appearance. As indicated by the yellow arrows, UL6 is missing in (a); both UL7 and UR7 are missing in (b); separators are located between multiple teeth (e.g., UR6 and UR7) in (c); UR6 and UL6 have braces in (d).

missing teeth and two cases with dental separators/braces. It can be seen that, in some cases, MeshSegNet can identify missing teeth (e.g., UL6 in Fig. 10 (a)) and additional separators (e.g., between UR6 and UR7 in Fig. 10 (c)), even when the training set did not include similar subjects. However, in some more challenging cases (e.g., Fig. 10 (b) and (d)), our method failed to properly handle missing teeth and additional braces. To further improve the robustness of MeshSegNet, including more extreme subjects as training samples is necessary. *Second*, in our current implementation, MeshSegNet was followed by an offline post-processing step based on graph cuts for results refinement, which is sensitive to tuning parameters in practice. Therefore, integrating a fully trainable post-processing module (e.g., based on RNN and conditional random fields [36]) or capsule modules (such as in SegCaps [37] and 3D-PointCapsNet [38]) into our end-to-end MeshSegNet for enhancing the modeling of part-to-whole relationship on dental surfaces should be an interesting research direction in the future. *Third*, in our current implementation, different raw features (i.e., vertex coordinates, cell normal, and cell relative position) were simply concatenated as a single vector for network training. Regarding them as the initial description of the same target from different views (such as in [23], [39]) to construct a multi-view MeshSegNet could potentially maximize the contributions of different inputs to learn more discriminative geometric features for surface labeling.

VI. CONCLUSION

In this work, we have proposed to automatically label individual teeth on raw dental surfaces acquired by 3D intra-oral scanners. An end-to-end deep neural network (called MeshSegNet) has been proposed, which learns hierarchically multi-scale contextual features and holistic features directly

from raw surface input, and then applies a dense fusion strategy to combine local-to-global geometric features for mesh cell labeling. A post-processing step based on graph cuts has been used to refine the results for MeshSegNet. Evaluating MeshSegNet on a real-patient dataset, the results show that it outperforms the state-of-the-art methods for 3D shape segmentation with a large margin.

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